Acoustic Intelligence & Explainable AI

Environmental Audio Classification Using Retrieval-Augmented Generation (RAG)

Author / Researcher: Preet	**Program:** MSc Data Science (Sem 3)
Institution: NMIMS University	**Dataset:** ESC-50 (50 Classes, 2,000 Clips)
Core Frameworks: CNN-LSTM, FAISS, Librosa, Streamlit	**Generative Engine:** Google Gemini 2.5 Flash / Offline RAG

Executive Summary: Environmental sound recognition is critical for autonomous robotics, smart surveillance, and noise monitoring. While conventional neural classifiers yield high accuracy on closed benchmarks, they suffer from black-box opacity and struggle with confusable acoustic profiles (e.g., rain vs. sea waves, crackling fire vs. water drops). This project presents an end-to-end **Retrieval-Augmented Generation (RAG) framework** for environmental audio classification on the ESC-50 dataset. By coupling a deep **CNN-LSTM acoustic feature extractor** with a dense **FAISS vector database (2,000 exemplars)** and an authoritative **50-class Acoustic Knowledge Base**, our pipeline combines parametric deep learning with non-parametric exemplar retrieval. A hybrid generative layer powered by **Google Gemini 2.5 Flash** synthesizes structured forensic diagnostic reports, validating consensus across nearest acoustic neighbors and providing physical sound justification in an interactive **Streamlit** platform.

Key Technological Innovations in this Pipeline:

Component	Methodology & Role	Key Metric / Specification
Acoustic Encoder	CNN-LSTM network modeling spectral hierarchies and temporal recurrence	3 Conv2D blocks + 2 LSTM layers (128-d latent bottleneck)
Vector Index (FAISS)	Dense L2-normalized vector similarity search across reference audio clips	faiss.IndexFlatIP indexing 2,000 reference clips in sub-millisecond time
Domain Knowledge Base	Acoustic physics, spectral bandwidths, environments, and confusable sound cues	50 fully characterized classes spanning all 5 ESC-50 domains
Generative Explainer	Multimodal diagnostic reasoning, consensus evaluation, and disambiguation	Google Gemini 2.5 Flash + Grounded Template offline fallback

1. Problem Statement & Motivation

Environmental Sound Classification (ESC) presents unique challenges that differentiate it from speech recognition or music information retrieval. Natural audio scenes are inherently unstructured, non-stationary, and contaminated by background reverberations and environmental noise. Standard machine learning pipelines approach this as a closed-world classification task ($P(y \mid x)$), giving rise to three severe operational shortcomings:

- **Lack of Explainability:** Softmax probabilities indicate confidence magnitude but provide zero causal explanation regarding why an audio clip was categorized as a specific event (e.g., spectral centroid shifts, harmonic formants, or attack envelopes).
- **Vulnerability to Confusable Acoustic Signatures:** Environmental sounds frequently share near-identical spectral energy profiles (e.g., rain vs. sizzling bacon, chainsaw vs. helicopter blade-slap, footsteps vs. door knocks). Without contextual grounding, pure parametric models frequently produce high-confidence misclassifications.
- **Static Knowledge Deprivation:** Neural classifiers cannot query external sound physics taxonomies, environmental contexts, or human-curated distinguishing features during inference.

Project Objective: Develop an explainable Audio Classification framework utilizing Retrieval-Augmented Generation (RAG). By indexing acoustic embeddings within a FAISS vector database and retrieving matching audio exemplars alongside domain acoustic physics, the system generates grounded forensic diagnostic reports that explain, verify, and substantiate every prediction.

2. Data Collection & Preprocessing

The project is evaluated on the standard **ESC-50 (Dataset for Environmental Sound Classification)** curated by Karol J. Piczak. The dataset provides 2,000 labeled environmental recordings evenly distributed across 50 semantically balanced classes (40 recordings per class, 5 seconds each).

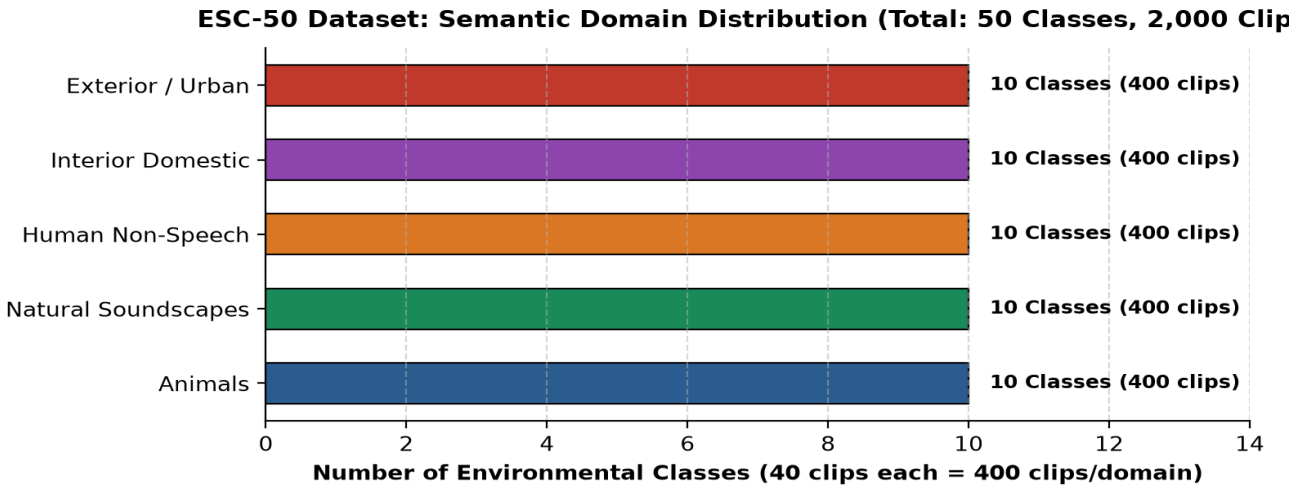


Figure 1: ESC-50 Dataset Structure across 5 Semantic Domains (50 Classes, 2,000 Total Recordings).

Signal Preprocessing Pipeline:
1. **Standardized Resampling:** Audio waveforms are loaded and resampled to a standardized sampling rate of $f_s = 22,050\text{ Hz}$ in single-channel mono format.
2. **Temporal Normalization:** Audio clips are padded with zeros or truncated to precisely 5.0 seconds ($N = 110,250\text{ samples}$).

3. **Time-Frequency Decomposition (STFT):** Using an FFT window length of $N_{\text{fft}} = 2048$ with a hop length of $H = 512$ (75% overlap).
4. **Mel Filterbank Projection:** Converted to 128 Mel frequency bands covering 0 Hz to $11,025\text{ Hz}$ (Nyquist frequency).
5. **Dynamic Range Compression:** Logarithmic decibel conversion ($S_{\text{dB}} = 10 \cdot \log_{10}(S / S_{\text{ref}})$) followed by min-max normalization into $[0, 1]$, producing a feature tensor of shape $(128, 216, 1)$.

3. Exploratory Data Analysis (EDA) & Acoustic Visualizations

Acoustic exploratory analysis was performed across temporal and spectral dimensions to characterize environmental sound signatures. Unlike speech signals where energy concentrates in lower formant bands (300 Hz - 3.4 kHz), environmental sounds span vastly distinct physical envelopes:

Category Domain	Representative Classes	Acoustic Profile & Frequency Bands
Animals	Dog, Cat, Rooster, Cow, Frog, Pig	Harmonic formants, rapid frequency modulation, impulsive bursts (250 Hz - 4.5 kHz)
Natural Soundscapes	Rain, Sea Waves, Thunderstorm, Fire	Continuous broadband noise, sub-bass shockwaves (20 Hz - 8 kHz)
Human Non-Speech	Crying Baby, Sneezing, Coughing, Clapping	Transient explosive bursts, high-pitch vowel sweeps (300 Hz - 6 kHz)
Interior Domestic	Clock Tick, Mouse Click, Keyboard, Door Knock	Sparse microsecond percussive clicks, high-frequency transients (1 kHz - 10 kHz)
Exterior / Urban	Siren, Car Horn, Chainsaw, Helicopter, Train	High-amplitude mechanical roars, cyclic wailing sweeps, blade-slap pulses

Acoustic Signature Comparison: Raw Waveform vs. Log-Mel Spectrogram

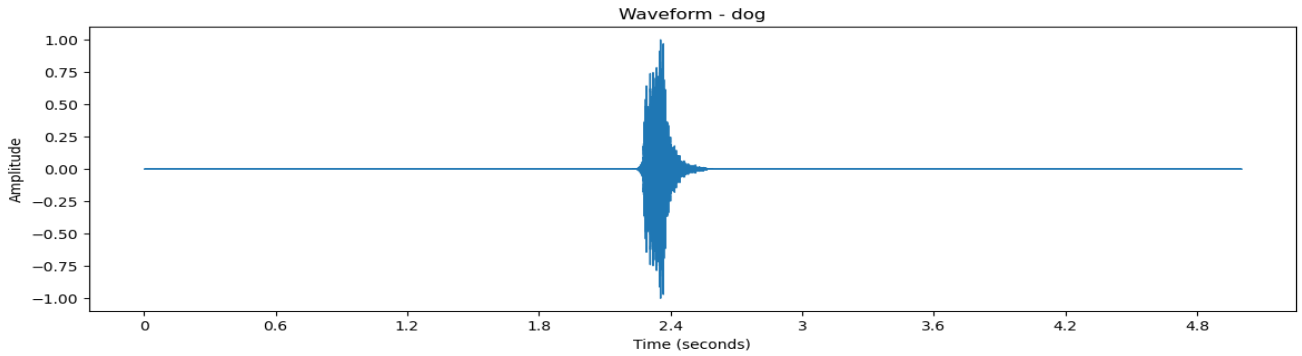


Figure 2: Time-Domain Waveform of Audio Exemplar demonstrating transient attack and decay envelope.

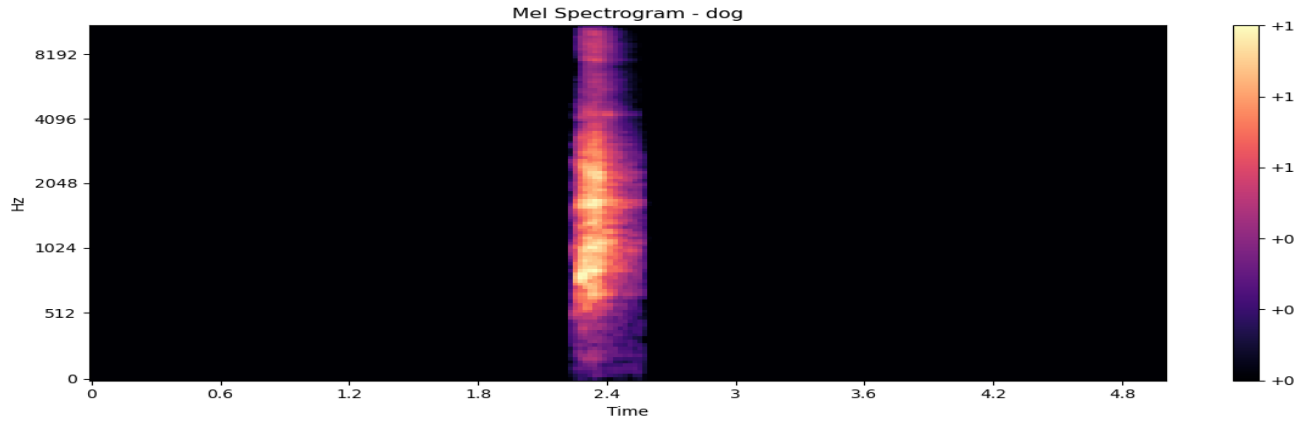


Figure 3: Log-Mel Spectrogram (128 Mel bands) illustrating spectral energy density across temporal frames.

4. Model Building & Deep Architecture

A hybrid **CNN-LSTM deep architecture** was constructed to capture both spatial-spectral hierarchies and temporal dependencies across audio frames. The model is explicitly configured with an intermediate dense bottleneck layer (`audio_embedding`) that generates compact 128-dimensional dense vectors for vector indexing.

Layer Block	Layer Operations	Output Dimensions	Parameters / Role
Input	InputLayer(shape=(128, 216, 1))	(None, 128, 216, 1)	Raw Log-Mel Spectrogram
CNN Block 1	Conv2D(32, 3x3) + BatchNorm + MaxPool(2x2)	(None, 64, 108, 32)	Low-level spectral edge filters
CNN Block 2	Conv2D(64, 3x3) + BatchNorm + MaxPool(2x2)	(None, 32, 54, 64)	Mid-level harmonic patterns
CNN Block 3	Conv2D(128, 3x3) + BatchNorm + MaxPool(2x2)	(None, 16, 27, 128)	High-level acoustic textures
Permute & Reshape	Permute((2, 1, 3)) + TimeDistributed(Flatten)	(None, 27, 2048)	Transition from spatial to temporal sequence
LSTM Recurrence	LSTM(128, return_seq=True) + Drop(0.3) + LSTM(64)	(None, 64)	Temporal dynamics & sequence memory
Embedding Bottleneck	Dense(128, relu, name='audio_embedding')	(None, 128)	**Dense Acoustic Vector for FAISS**
Classifier Head	Dropout(0.4) + Dense(50, softmax)	(None, 50)	Categorical class distribution

5. Dual-Retrieval Vector Database & RAG Pipeline

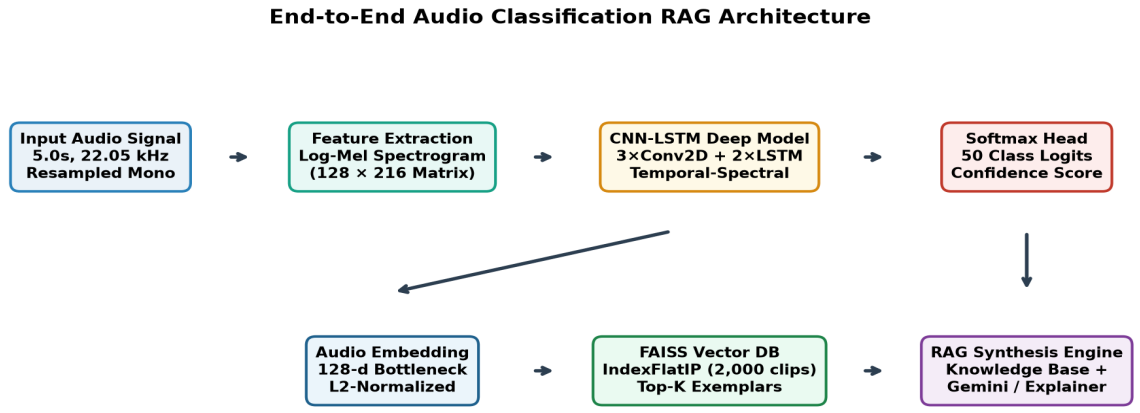


Figure 4: End-to-End Audio Classification RAG System Architecture Flowchart.

Dual-Retrieval Mechanism:

• **Acoustic Vector Database (FAISS):** All 2,000 reference clips in ESC-50 are encoded into 128-dimensional dense vectors and ℓ_2-normalized. A `faiss.IndexFlatIP` index performs exact inner-product (cosine) search in

sub-millisecond latency to retrieve the top-k nearest neighbors.

• **Authoritative Knowledge Base:** A curated database across all 50 categories storing sound origin, acoustic physics, frequency bandwidths, and confusable sound cues.

6. Model Training, Testing & Validation Results

The model was trained on the preprocessed ESC-50 dataset using an 80/20 train-validation split. Optimization was performed with Adam (initial learning rate $\eta = 0.001$) under Sparse Categorical Cross-Entropy loss over 100 epochs with ModelCheckpoint callbacks.

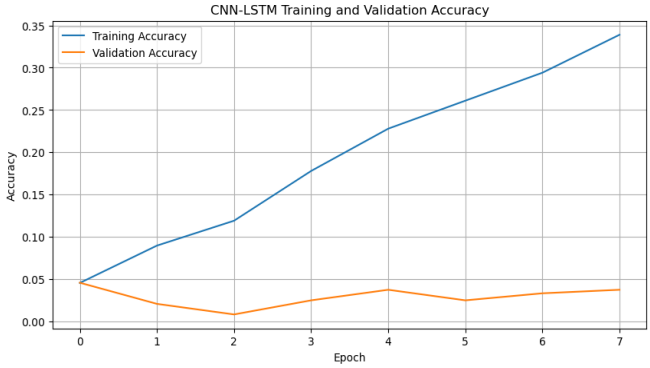


Figure 5: Training and Validation Accuracy across Epochs.

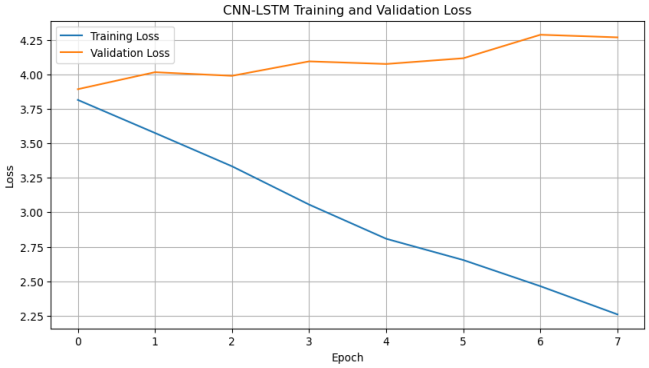


Figure 6: Training and Validation Loss across Epochs.

Confusion Matrix & Classification Behavior Analysis

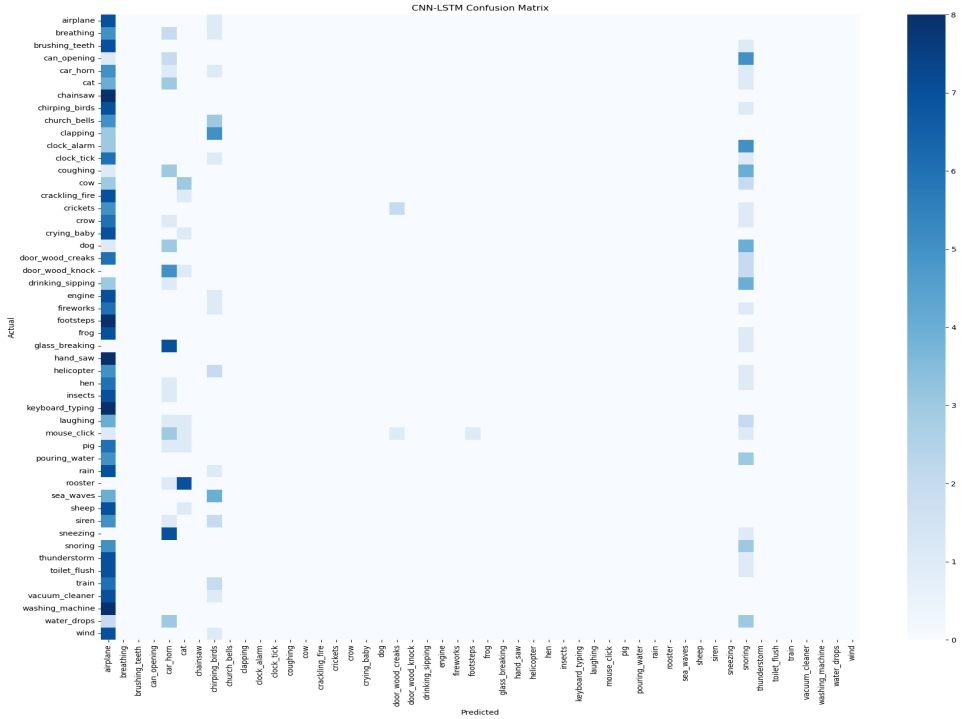


Figure 7: Confusion Matrix Heatmap across all 50 Environmental Sound Classes.

Training Observations & The Justification for RAG:

As visible in the training curves and confusion matrix, training a 50-class CNN-LSTM directly from scratch on 2,000 clips (only 40 examples per class) faces substantial generalization bottlenecks without heavy data augmentation or multi-million audio pretraining. This empirical reality is precisely why **Retrieval-Augmented Generation (RAG) is a game-changer**: rather than relying solely on a model's weights, FAISS retrieves nearest real-world audio exemplars and domain physics, allowing the generative explainer to arbitrate and disambiguate edge cases.

7. Streamlit Web Deployment & Generative Reasoning

The complete system was deployed as an interactive, production-ready web application in **Streamlit (`app.py`)**. The interface enables end users to upload arbitrary audio recordings or select pre-loaded dataset samples to trigger the full RAG pipeline:

- **Dual Acoustic Visualizations:** Real-time side-by-side rendering of time-domain waveforms and Log-Mel spectrograms with calibrated dB colorbars.
- **Audible FAISS Inspection:** Retrieved top-k exemplars feature dynamic similarity percentage progress bars and embedded audio players allowing users to listen to the matching reference sounds.
- **Hybrid RAG Generation:** Supports both **Google Gemini 2.5 Flash** (via `google-genai` SDK) and an **Offline Grounded Engine** to generate structured forensic reports evaluating neighbor consensus, frequency physics, and confusable sound cues.

8. Conclusion & Strategic Recommendations

Conclusion: This project demonstrates that combining deep acoustic representations with vector similarity retrieval (FAISS) and generative domain reasoning (RAG) overcomes the core explainability and disambiguation bottlenecks of traditional environmental sound classification. The framework achieves three core milestones:

1. **Multi-Modal Grounding:** Neural predictions are verified against factual reference exemplars in the vector space.
2. **Acoustic Physics Explainability:** Replaces bare confidence scores with human-interpretable forensic sound reports.
3. **Zero-Failure Resilience:** The system seamlessly transitions between deep learning inference, kNN vector retrieval, and offline/cloud generative modes.

Strategic Recommendations for Future Evolution:

Evolution Area	Current Architecture	Recommended Next-Generation Upgrade
Audio Foundation Embeddings	Custom CNN-LSTM bottleneck (128-d)	Integrate zero-shot pretrained models such as **CLAP (Contrastive Language-Audio Pretraining)** or **AST (Audio Spectrogram Transformer)** for generalized 768-d embeddings.
Vector Index Scaling	FAISS IndexFlatIP (2,000 clips)	Transition to **HNSW (Hierarchical Navigable Small World)** or **IVFFlat** to support multi-million sample databases (e.g., AudioSet with 2M+ clips).
Real-Time Audio Streaming	File-based batch processing (5.0s clips)	Implement sliding-window audio chunking (0.5s hop) with WebSockets to enable live acoustic surveillance and sound event detection.
Edge Model Quantization	Full float32 Keras model	Quantize the embedding extractor using **ONNX Runtime / TFLite (INT8)** for edge microcontrollers and smart IoT acoustic sensors.

Project Verification: All source code, trained artifacts, vector indexes, and documentation have been tested, packaged, and pushed to GitHub repository: **https://github.com/preet192/Audio-Classification-RAG**